



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

Chandigarh University Uttar Pradesh

School of Computer Science & Engineering

Semester III

Python Programming

Course Code: 25 CSH-214

Faculty Name: Prateek Singhal

Python

Course Objectives

Students learn basic syntax, data types, control structures, functions, and the use of standard libraries such as NumPy and math

Students will be able to solve real-time problem-solving.

By the end of the course, students develop the ability to write efficient programs, evaluate their correctness and performance.



CO-PO Mapping

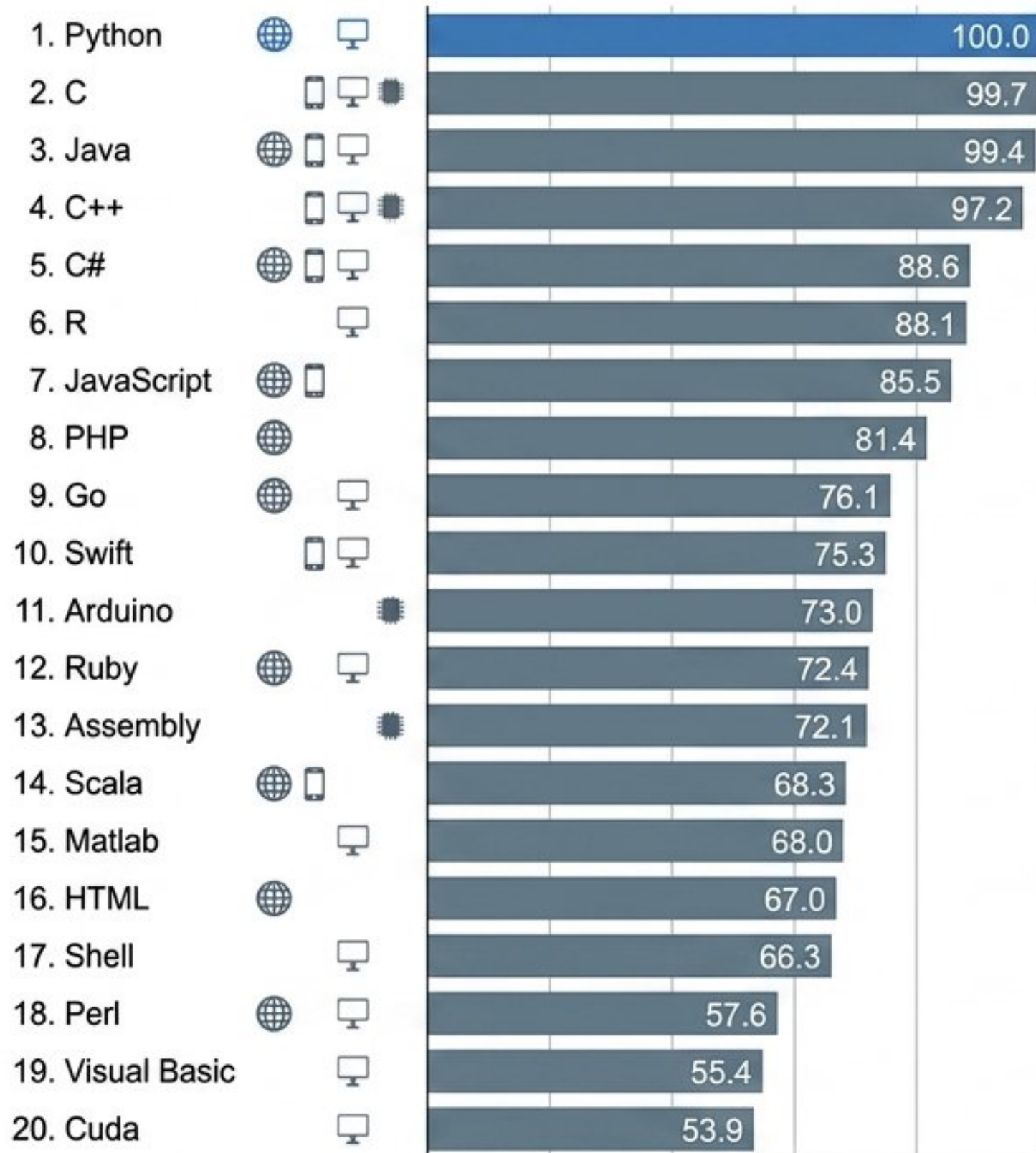
COs/ Pos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	-	-	2	3	2
CO2	3	3	2	2	3	-	-	-	-	-	-	2	3	3
CO3	3	3	2	2	3	-	-	-	-	-	-	2	3	3
CO4	3	3	2	2	3	-	-	-	-	-	-	2	3	3
CO5	3	3	3	3	2	3	2	2	3	2	2	3	3	3

UNIT 1

Topics to be Covered

- Writing a Python Program
- Basic Building Blocks of Programs
- Input
- Output
- Sequential Execution

The Undisputed Standard of Modern Code



The Zen of Python

"Beautiful is better than ugly."

"Explicit is better than implicit."

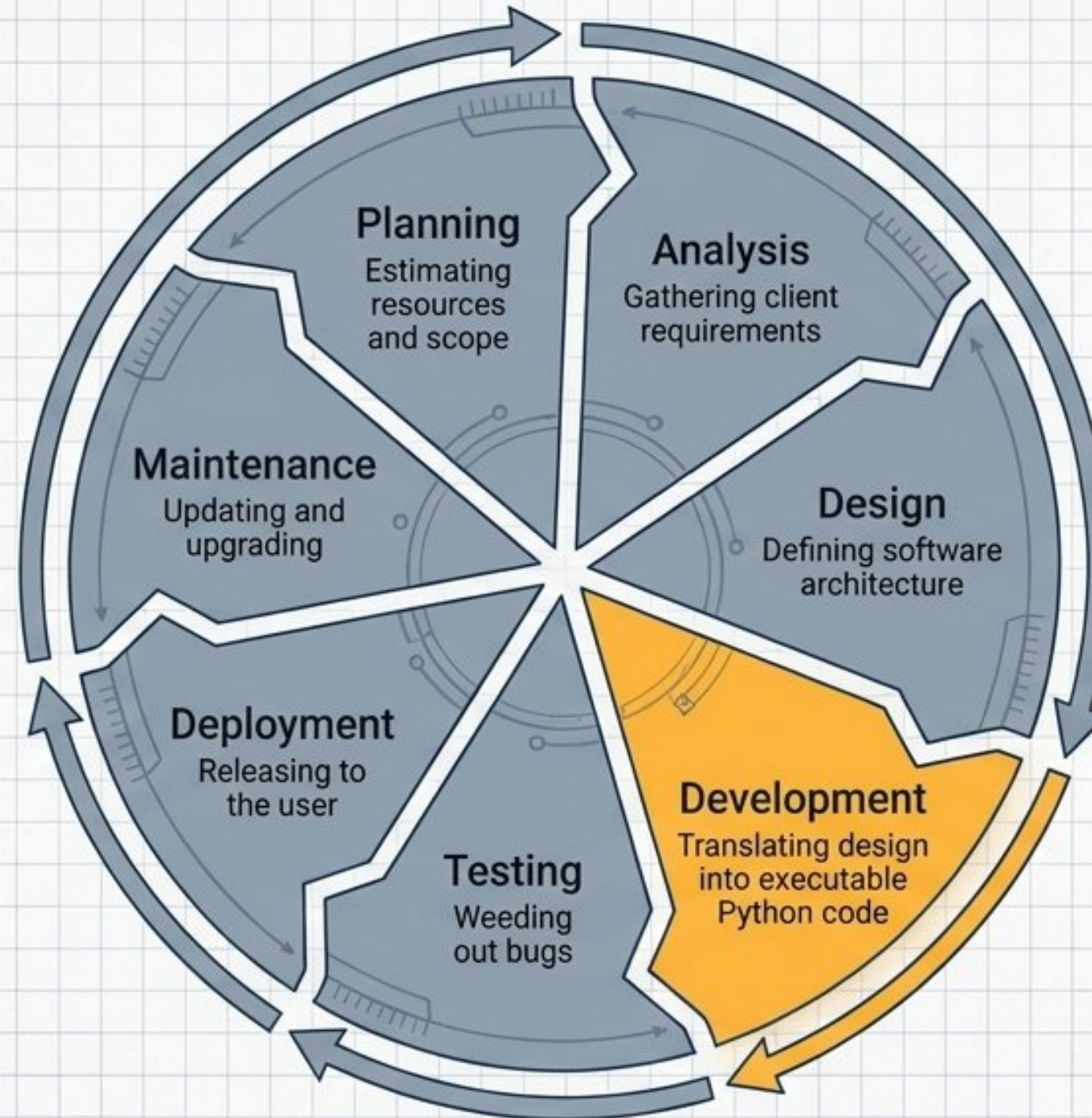
"Simple is better than complex."

"Readability counts."



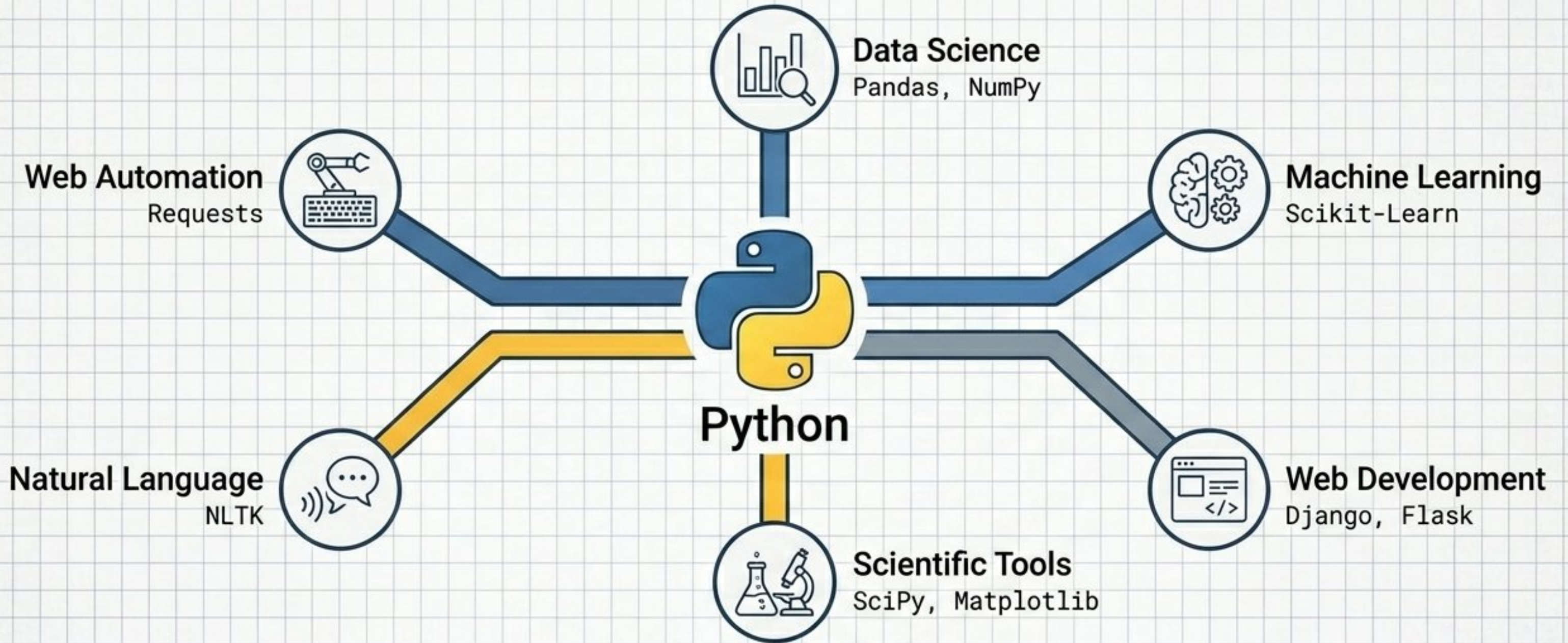
Python is guided by its creator, Guido van Rossum, known as the Benevolent Dictator For Life (BDFL).

The Software Development Life Cycle (SDLC)



Programming is just one crucial gear in a much larger machine.

One Language, Infinite Applications



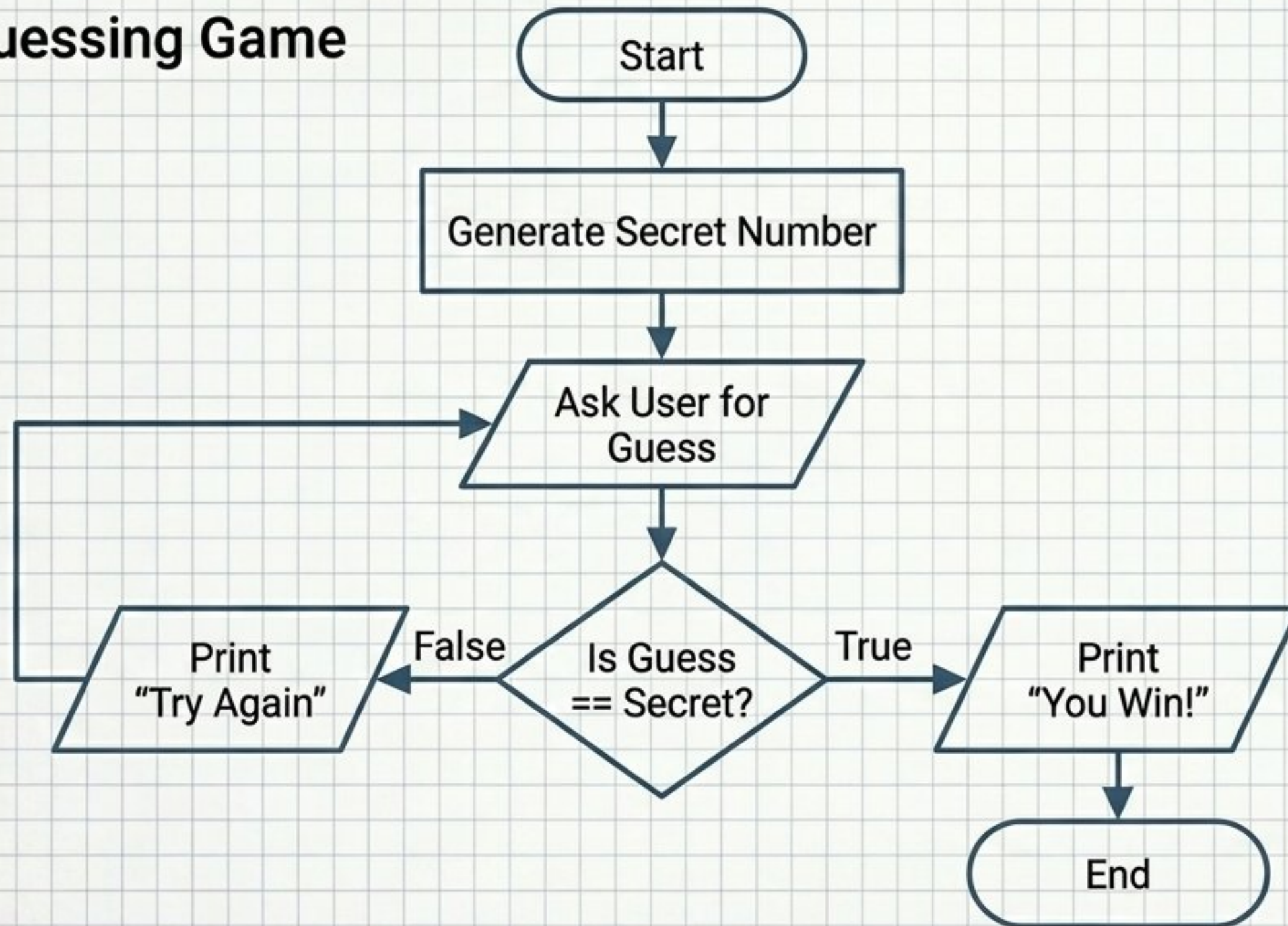
Python relies on a "batteries included" philosophy—a rich set of built-in libraries that speed up development.

The 2026 IDE Diagnostic Matrix

Tool	Key Metaphor	Best For	Cost		The Verdict
VS Code	Super-powered notepad	General Purpose & Web	Free	●	The industry standard (requires plugin setup).
Cursor	AI-powered VS Code	AI-Assisted Coding	Freemium	●	The future; chats with your codebase.
PyCharm	Fully assembled factory	Large Pro Projects	Freemium	●	Heavyweight champion; zero setup required.
Jupyter / Colab	Live interactive notebook	Data Science	Free	●	Essential for data; mixes code and visualizations.
Thonny	Stripped-down workshop	Total Beginners	Free	●	Best for Day One; excellent variable visualization.

Logic Before Syntax: Architectural Schematics

Guessing Game



Flowcharts are a structured graphical way to document process flow.

A computer program is just branching decisions and repeating loops translated into text.

The Raw Materials: Anatomy of a Script

Variables

Containers labeling memory locations.

Data Types

Integers and Floating-point numbers.

Operators

Manipulating data just like in math.

f-strings

Formatting text outputs dynamically.

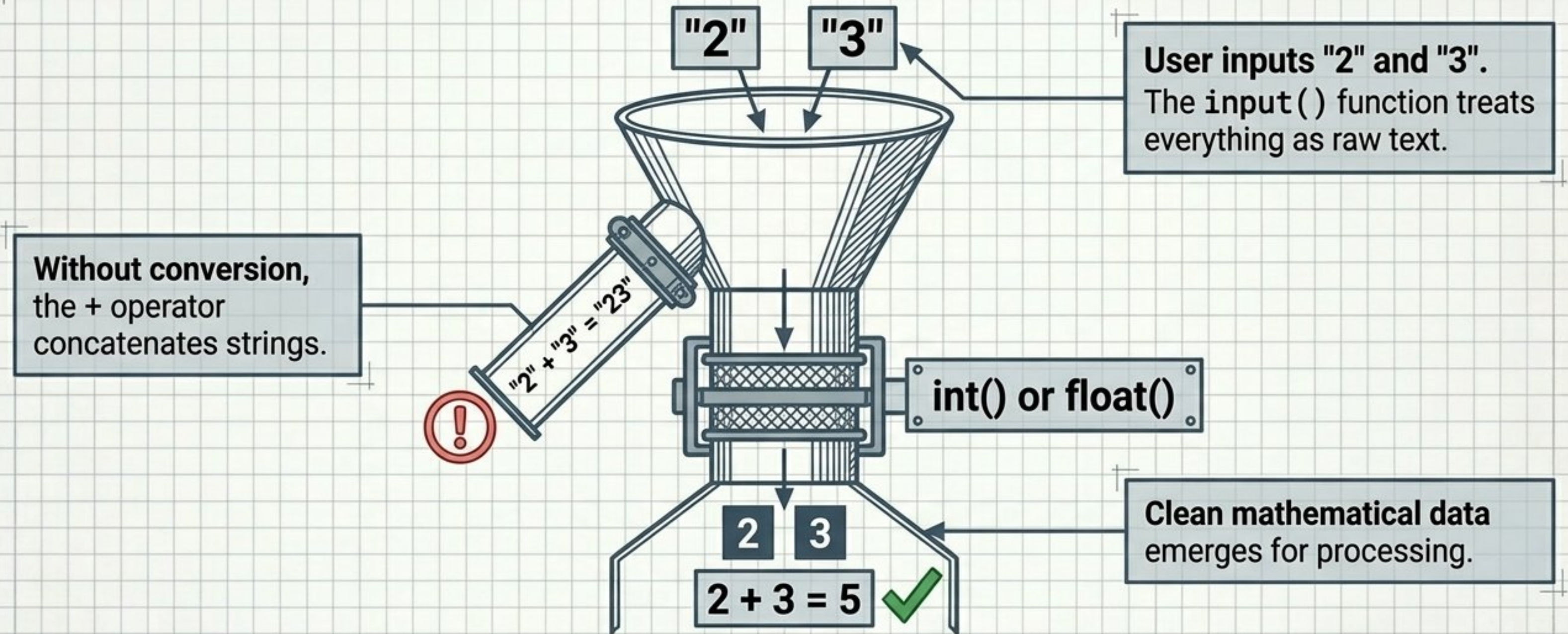
Functions

Reusable blocks of code performing a specific task.

```
weight = 75
height = 1.80
bmi = weight / (height * height)
print(f'Your BMI is {bmi}')
```

The diagram illustrates the components of a Python script using a sample code block. A central box contains the code: `weight = 75`, `height = 1.80`, `bmi = weight / (height * height)`, and `print(f'Your BMI is {bmi}')`. Arrows point from five surrounding boxes to specific parts of the code: 'Variables' points to `weight` and `height`; 'Data Types' points to the values `75` and `1.80`; 'Operators' points to the `/` and `*` symbols; 'f-strings' points to the `f'Your BMI is {bmi}'` string; and 'Functions' points to the `print` function call.

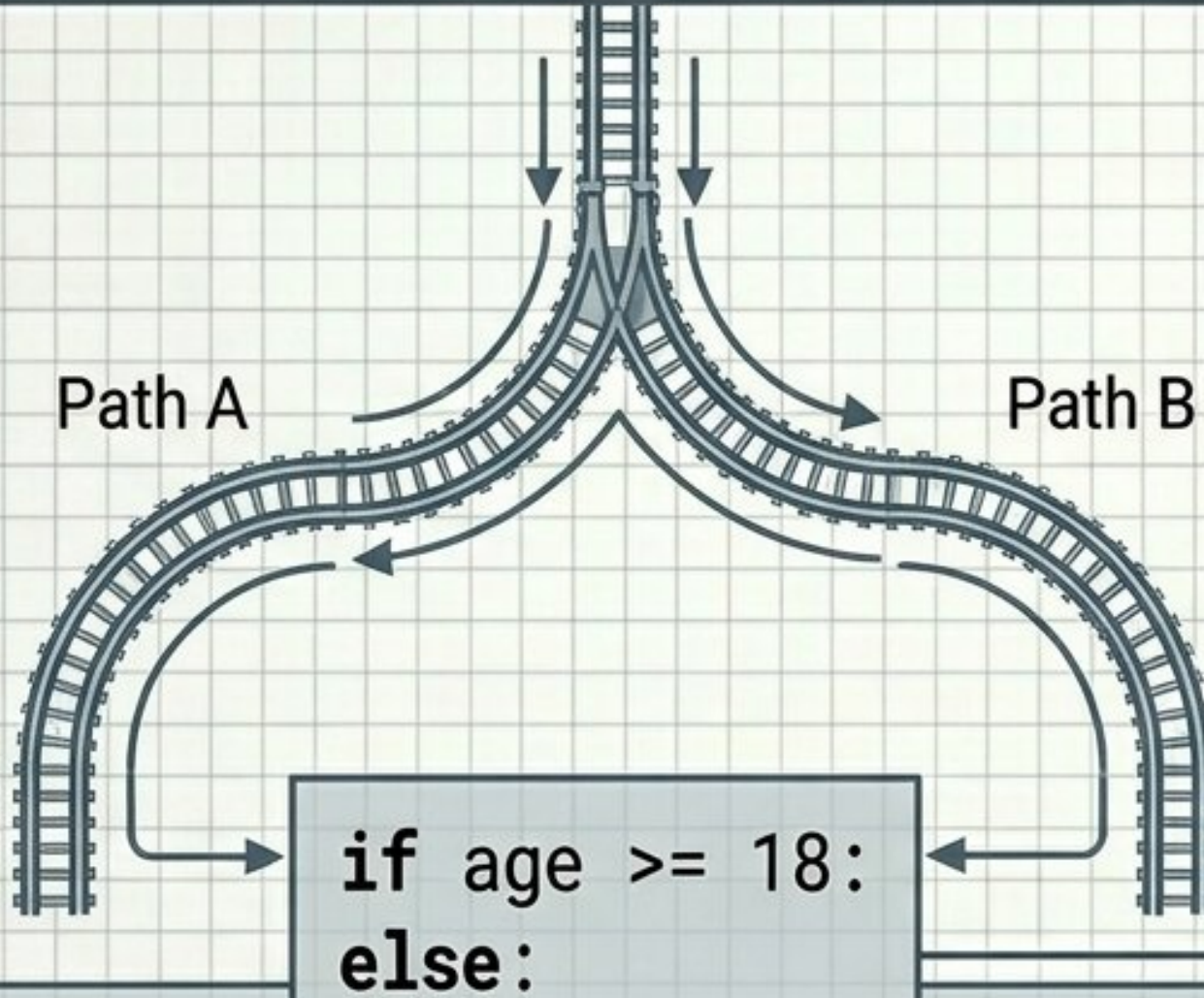
The Type Conversion Funnel



A computer only knows what you explicitly tell it. Text and Math are physically different materials in the workshop.

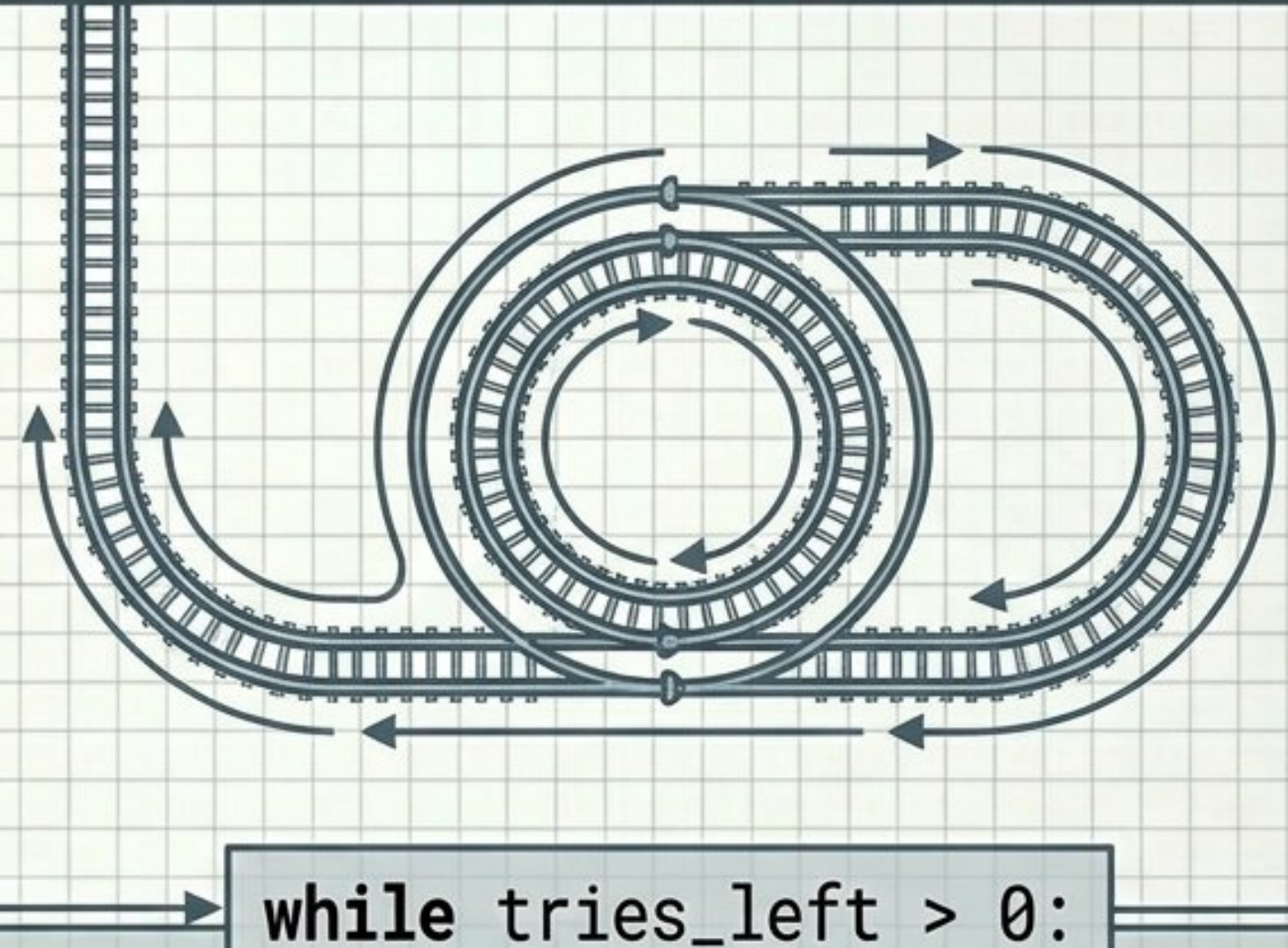
Directing Traffic: Control Structures

Branching (if / elif / else)



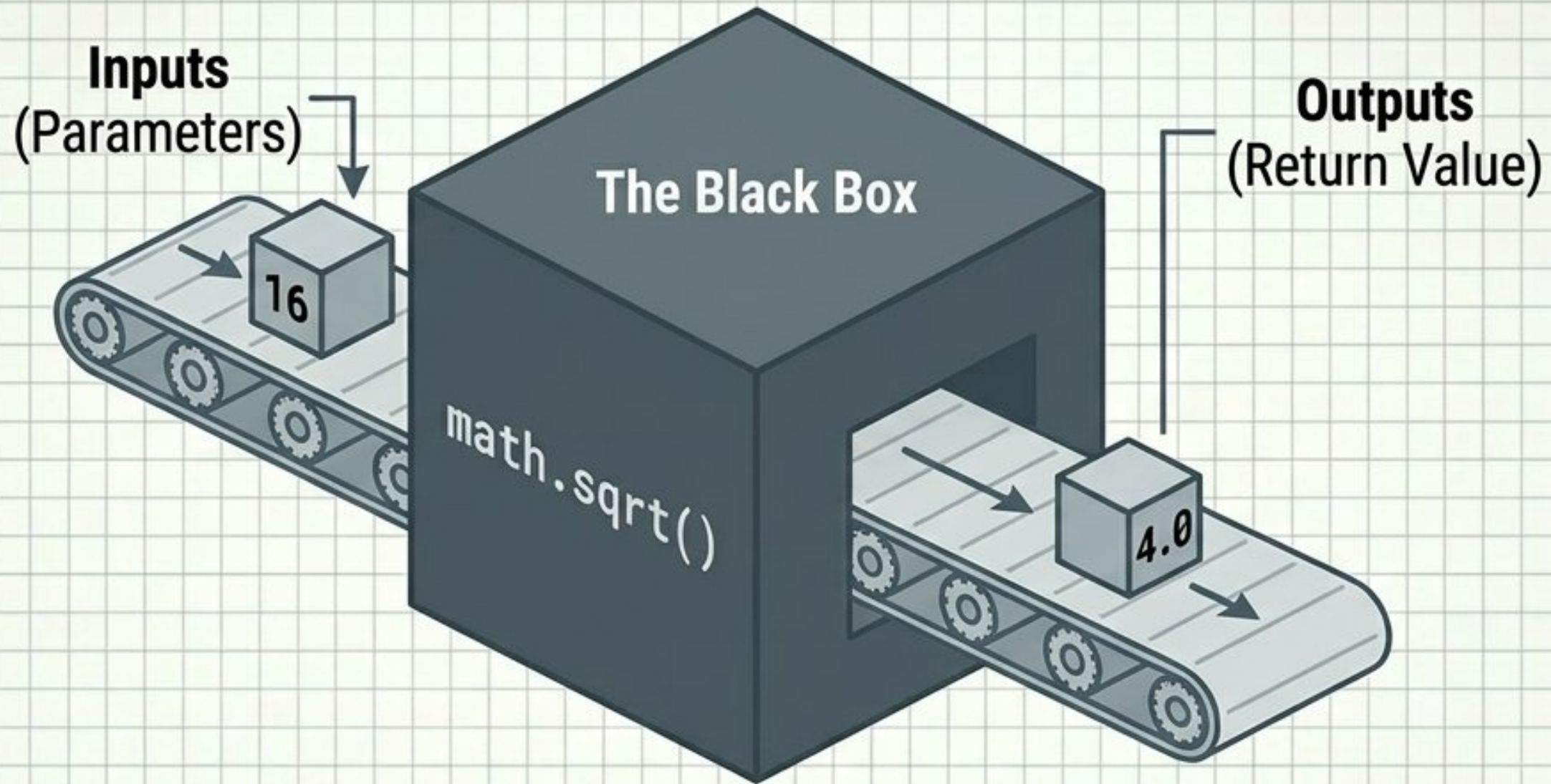
Manages the flow of a program by allowing conditional execution based on true/false checks.

Looping (for / while)



Repeats operations dynamically. Software is perfect for executing repetitive tasks efficiently.

Modular Machinery: The Black Box Principle



A function is a self-contained block of code. You don't need to know how the engine works inside to use it. You only need to know what data it requires and what result it hands back.

Note: Python's standard library provides thousands of these built-in 'black boxes' ready to use.

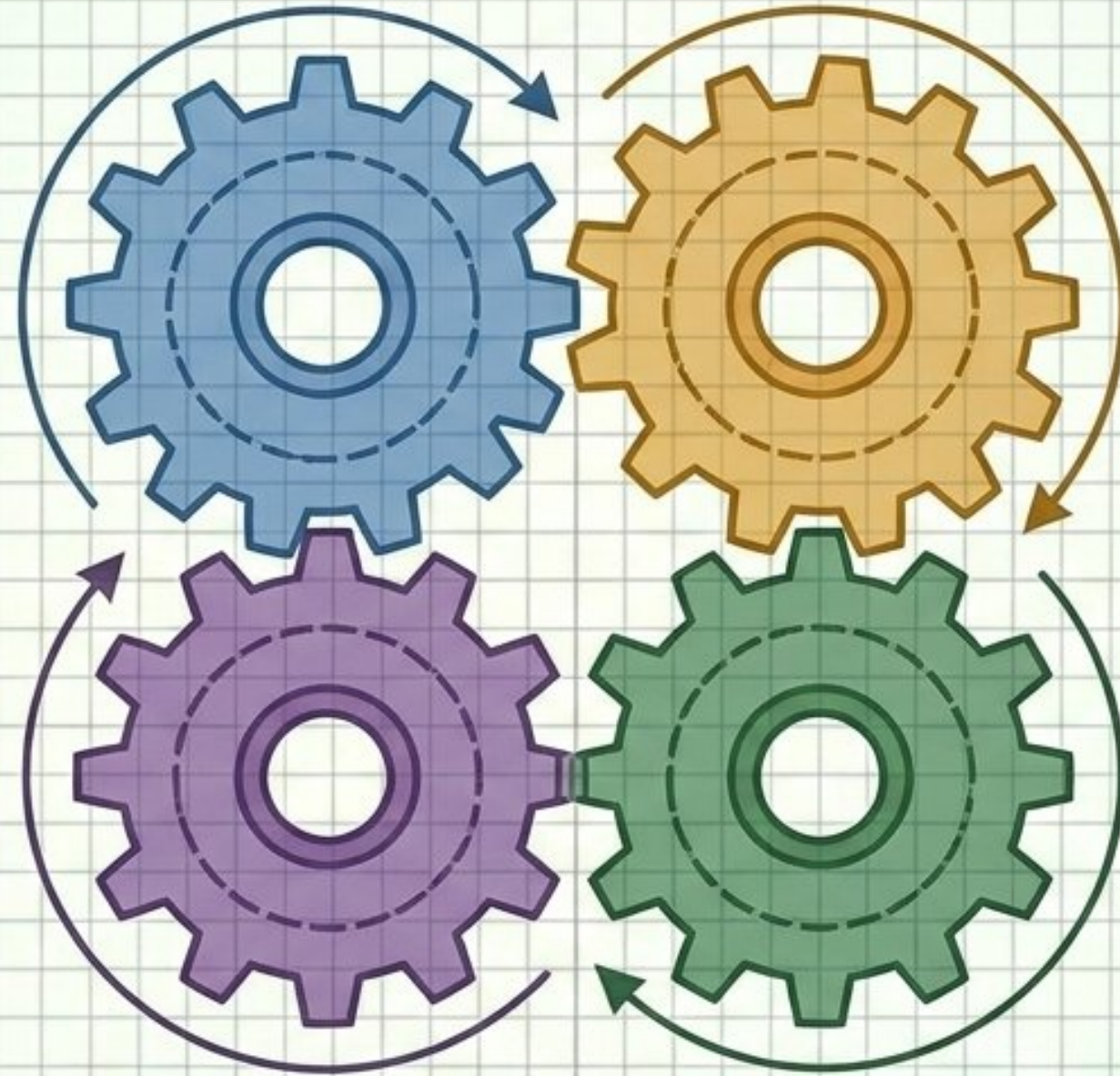
The Interpreter Engine: REPL

Read
Captures your input.

Loop
Resets the prompt
(`>>>`) and waits.

Eval
Crunches code
& calculates.

Print
Hands back
immediate feedback.



Python isn't compiled all at once. The interpreter reads source code one statement at a time. The REPL provides an interactive live-chat with your computer—perfect for testing small snippets on the fly.

The Execution Modes Panel

The Command Line (CLI)

```
$ python script.py
```

Executes a saved file from top to bottom.
Requires explicit `print()` statements to see output.

Interactive REPL

```
>>> 2 + 2  
4  
>>>
```

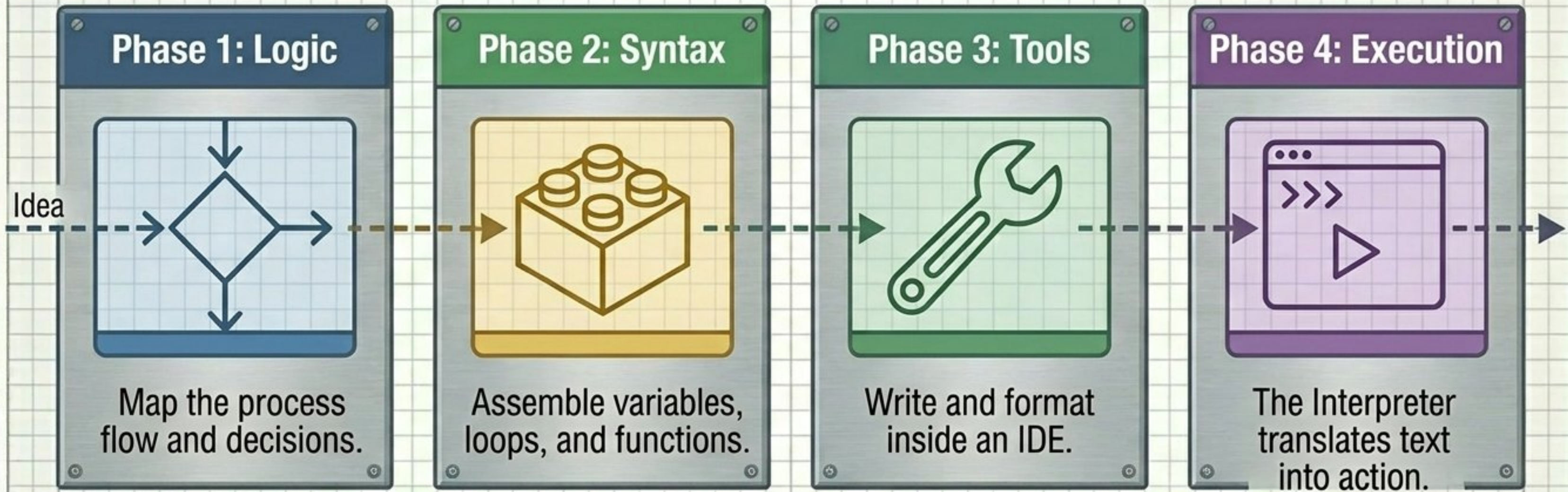
Launched by typing 'python'. Code runs instantly but vanishes when closed. Not persistent.

The IDE



The modern approach.
Edits, manages, and executes persistent scripts with a single click inside inside a unified workshop.

The Developer's Journey: Concept to Execution



Programming is not just typing syntax. It is the structured translation of human intent into machine execution.